

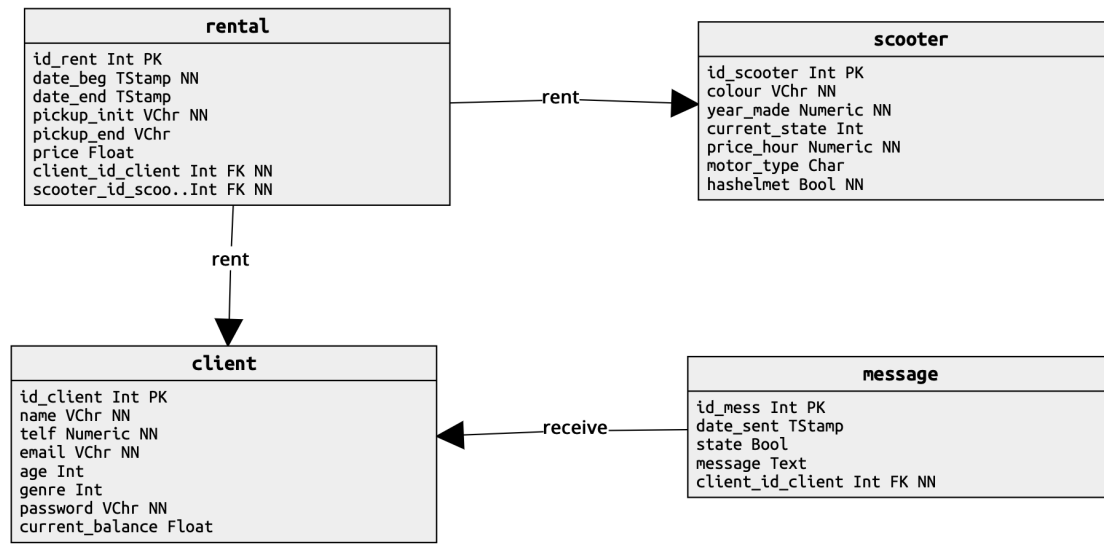


Data Analytics Technology

Exam questions examples

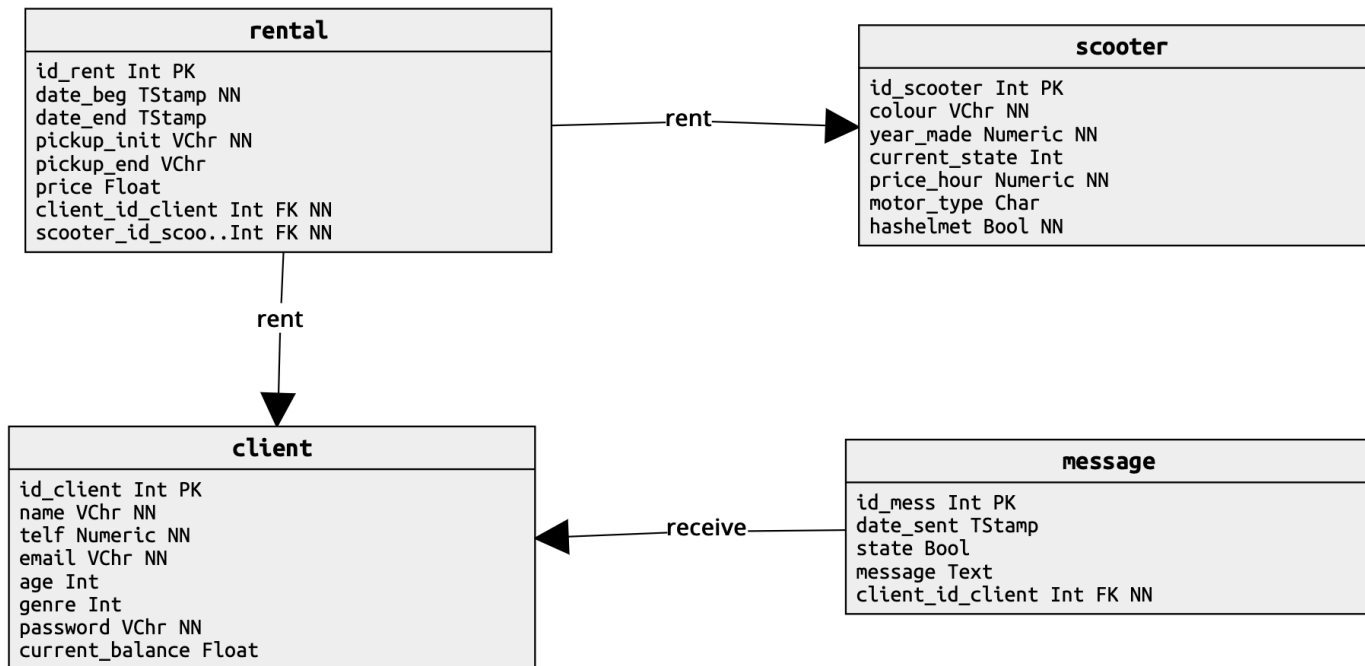
Catarina Silva, catarina@dei.uc.pt

Consider an electric scooter company supported by the following physical diagram.



- Electric scooters may be rented and returned by registered clients that use the company's mobile app in one of the several pickup and return points in the city. To make this possible, clients should charge their account previously and have enough balance (current_balance) for at least one hour in the specific scooter. Each scooter has a specific motor that can be of two types (type: T1 or T2) and may or may not include a box for the helmet (hasHelmet: True or False).
- The scooter company may also send messages to clients, either specific messages regarding their rentals or general information, e.g., promotions or new rates. The type of message is stored (type=private, type=general). When a client reads a message, it is registered in the message details (state=true).

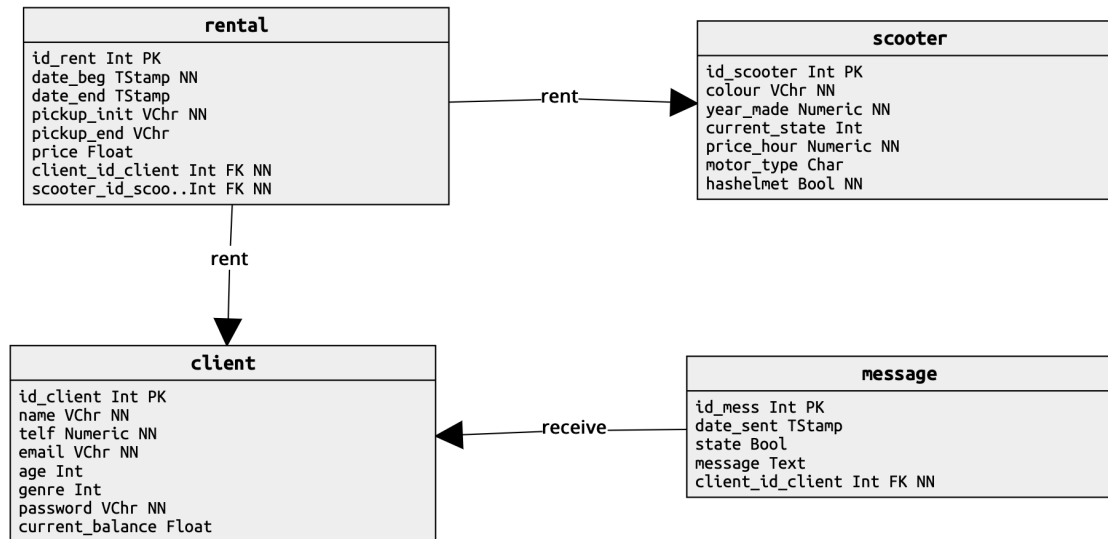
- The administrators of the scooter renting company intend to obtain strategic information for decision support from the data in its operational database for instance for determining the most solicited scooter types or the busiest pickup points. For this they decided to order a Data Warehousing and OLAP solution. This solution should enable the following four (4) questions to be answered or the requested information be presented, while at the same time promoting the ad-hoc exploitation of the data for the discovery of "new knowledge":
 1. Which are the busiest pickup points in terms of number of rentals and revenue? How does this evolve over the day/week/year?
 2. Is there a significant difference between men and women clients, namely in terms of rental time, scooter types (motor and helmet) and rental duration?
 3. Does age play a role in the rentals, namely regarding price and rental duration?
 4. Is there a relation between the reading of promotion messages (type=general) by the clients and their rentals?
- (a) Present a model for the Data Warehouse, not forgetting to make clear all **the facts, dimensions, granularity**, and all aspects relevant to the Data Warehouse project.



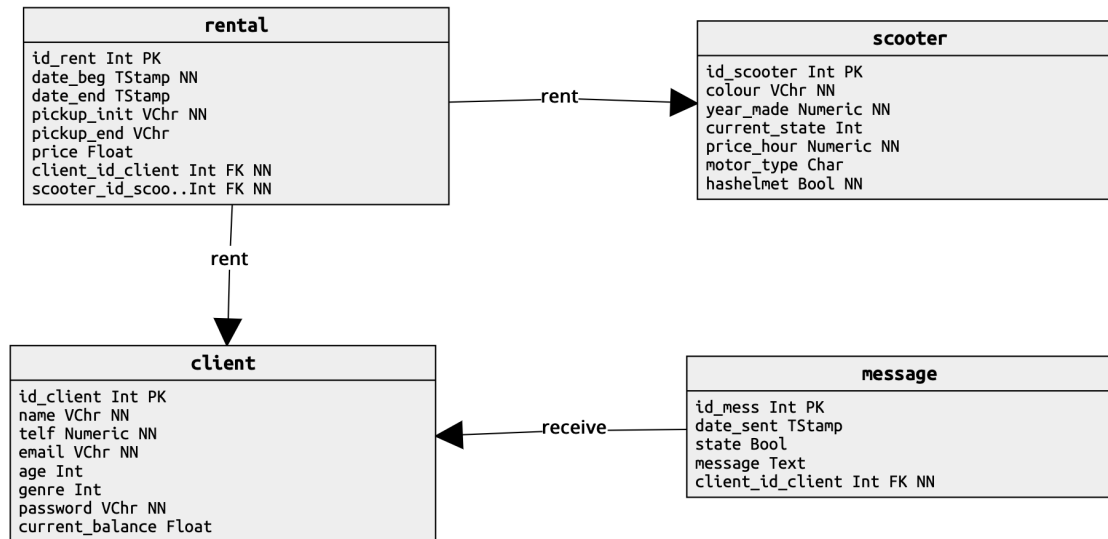
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(b) Explain how the previous model can address the previous 4 questions. (You can use text and/or SQL code.)

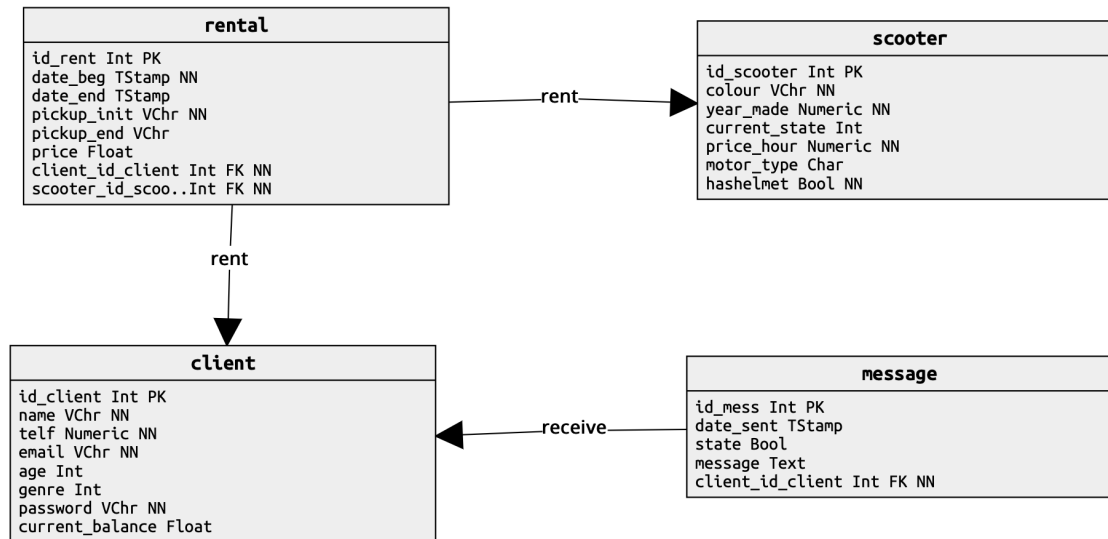
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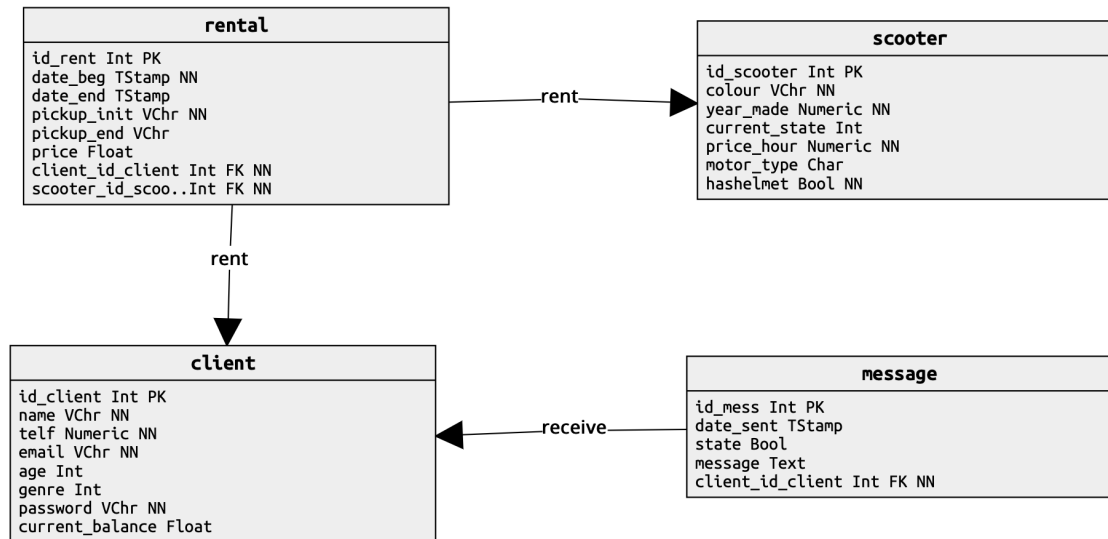
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(c) Considering these 4 questions (queries) which indexes would you create? Justify.